Lecture 14

The transfer function of a general second order system can be given as below :

$$H(s) = \frac{\omega_n^2}{\underbrace{1}_{a} \cdot s^2 + \underbrace{2\zeta\omega_n}_{b} s + \underbrace{\omega_n^2}_{c}}$$

The denominator is set equal to zero to find the poles. As it is a second order system it has two poles. The poles can be fond as

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2\xi\omega_n \pm \sqrt{(2\xi\omega_n)^2 - 4 \cdot 1 \cdot \omega_n^2}}{2}$$

$$= -\xi\omega_n \pm \sqrt{\xi^2\omega_n^2 - \omega_n^2}$$

$$= -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$$

if $\xi^2 - 1 < 0$, then the poles are complex.

$$\xi^2 - 1 < 0$$

$$\Rightarrow \xi^2 < 1$$

$$s_{1,2} = -\xi\omega_n \pm \omega_n\sqrt{-1(1-\xi^2)}$$

$$= -\xi\omega_n \pm \omega_n\sqrt{-1}\sqrt{1-\xi^2}$$

$$= -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

where $\sqrt{-1} = j$ is a complex number.

As a result, we have two complex roots of the form

$$s_{1,2} = -\sigma \pm j\omega_d$$

where $\sigma = \xi \omega_n$

$$\omega_d = \omega_n \sqrt{1-\xi^2} \qquad |\xi| < 1$$

We have found two poles of a second order system. These poles are complex. Additionally, we have expressed the poles in terms of $\boxed{\text{damping ratio } \xi}$ and $\boxed{\text{natural frequency } \omega_n}$.

ω_d = damped natural frequency